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Stabilization of lithium superionic conduction phase and enhancement of conductivity of LiBH_4 by LiCl addition

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LiBH_4 exhibits lithium superionic conduction accompanied by structural transition at around 390 K. Addition of LiCl to LiBH_4 drastically affects both the transition and electrical conductivity: Transition from low-temperature (LT) to high-temperature (HT) phases in LiBH_4 is observed at 370 K upon heating and the HT phase can be retained at 350–330 K upon cooling. Further, the conductivity in the LT phase is more than one or two orders of magnitude higher than that of pure LiBH_4 . These properties could be attributed to the dissolution of LiCl into LiBH_4 , suggested by *in situ* x-ray diffraction measurement. © 2009 American Institute of Physics.

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We have recently found the lithium superionic conduction in lithium borohydride (LiBH_4) accompanied by structural transition,¹ on the way to clarify the microwave absorbing mechanism of LiBH_4 .^{2,3} The electrical conductivity measured by ac complex impedance method jumped by three orders of magnitude due to structural transition from orthorhombic [low-temperature (LT)] to hexagonal [high-temperature (HT) superionic conduction] phases at approximately 390 K. The HT phase exhibited a high conductivity of the order of 10^{-3} S/cm. The conductivity calculated from the Nernst–Einstein equation using ^7Li -NMR correlation times was in good agreement with that measured by the impedance method. The lithium superionic conduction in LiBH_4 could potentially aid the development of solid electrolytes for all-solid-state batteries.^{4–8} Therefore, research directions on LiBH_4 are to stabilize the HT phase to least possible temperatures, and also to enhance the conductivity. It has been reported in various lithium ionic conductors that the addition of alkali metal halides improves their thermal stability and conductivity.^{9–11}

In this letter, we describe the effects of LiCl addition to LiBH_4 both on stabilization of the HT phase and enhancement of the electrical conductivity in the LT phase. The ad-

dition of LiCl to LiBH_4 was carried out by mechanical milling. Approximately 500 mg mixtures of LiBH_4 and LiCl (both from Aldrich Co. Ltd.) with molar ratios $x=0.33$ and 1 in $\text{LiBH}_4+x\text{LiCl}$ were mechanically milled for 5 h under Ar. The electrical conductivity was determined by ac complex impedance method between 303 and 433 K in heating and cooling runs.¹ Li foils were used as electrodes. The samples were examined by *ex situ* and *in situ* powder x-ray diffraction measurements (XRD) ($\text{Cu } K\alpha$ radiation) under Ar or He, and lattice parameters were determined by the Rietveld analysis using the RIETAN-2000.¹² Thermal analysis was carried out by differential scanning calorimetry (DSC) under He.

Prior to the study of the effects of LiCl addition on the electrical conductivity of LiBH_4 , only the milling effect was experimentally clarified. Although a single arc in the complex impedance plot has already been reported in the LT phase of pure LiBH_4 ,¹ two overlapping high- and low-frequency arcs are observed in the “as-milled” LT phase, as shown in the inset of Fig. 1(a), indicating the responses arising both from the bulk (capacitance (C):100–300 pF, corresponding to that for pure LiBH_4) and grain boundary (C :10–20 nF). In contrast, only a single arc corresponding

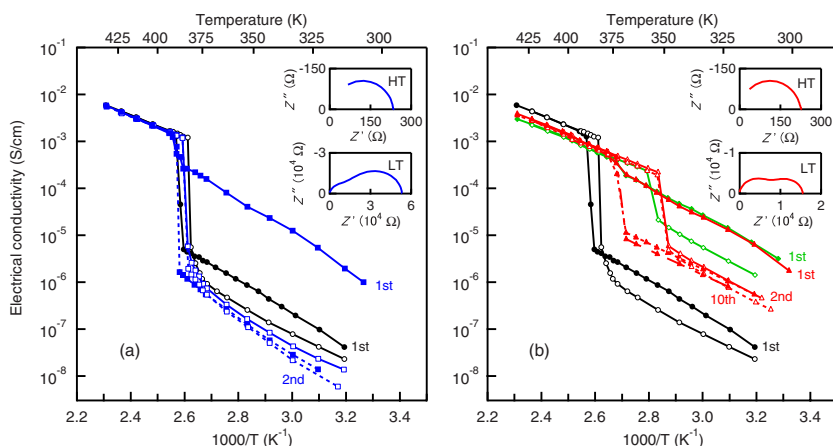


FIG. 1. (Color online) Temperature dependences of the electrical conductivity of (a) milled LiBH_4 (blue square) and (b) milled $\text{LiBH}_4+x\text{LiCl}$ [$x=0.33$ (red triangle) and 1 (green rhombus)]. Data of pure LiBH_4 (black circle) is also shown for comparison in both (a) and (b). Solid, dashed, and chain lines represent data measured in the 1st, 2nd, and 10th runs, respectively. Close and open symbols denote data taken in heating and cooling runs, respectively. Insets show the complex impedance plots taken in the 1st heating run for the LT and HT phases of (a) milled LiBH_4 and (b) milled $\text{LiBH}_4+0.33\text{LiCl}$.

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to the bulk conductivity is observed in the HT phase of milled LiBH_4 . We also confirmed that both the LT and HT phases exhibited single arcs in the second run. While the conductivity in the as-milled LT phase is two orders of magnitude higher than that of pure LiBH_4 , the temperature dependences of the conductivities after the LT-HT transition at around 390 K upon heating become almost the same as that of pure LiBH_4 , as shown in Fig. 1(a). Therefore, the effect of milling on conductivity, that is, a higher conductivity in the LT phase, was limited in the “as-milled (before heating)” LT phase.

Next, the electrical conductivities of milled $\text{LiBH}_4 + x\text{LiCl}$ ($x=0.33$ and 1) were investigated, and the results are shown in Fig. 1(b). For both $x=0.33$ and 1, the conductivities in the as-milled LT phases are approximately three times higher than that of milled LiBH_4 , and the HT phases show as high conductivity as pure and milled LiBH_4 . The lower conductivities in the LT phases after the HT-LT transition upon cooling as compared to those in the as-milled LT phase is predicted to be the same feature as mentioned above. Only the as-milled LT phases indicate the bulk and grain boundary conductivities, as shown in the insets of Fig. 1(b) (here we show the result for $x=0.33$).

Here, it should be emphasized that the LiCl addition drastically affects both the structural transition and electrical conductivity as follows: (1) the LT-HT transition is already observed at 370 K upon heating and the HT phase can be retained at 350 K upon cooling. Hysteresis becomes about 20 K by the LiCl addition (for $x=0.33$), as opposed to 4 K for pure and milled LiBH_4 . (2) The conductivity in the LT phase after the HT-LT transition upon cooling is still more than one or two orders of magnitude higher than those of pure and milled LiBH_4 .

These notable properties, that is, the stabilization (lower transition temperatures and larger hysteresis) of the HT phase and the enhancement of the electrical conductivity in the LT phase, are predicted to dominantly correlate with the dissolution of LiCl into LiBH_4 . Thus, the dissolution phenomenon was thoroughly investigated by *in situ* XRD, and the results are summarized in Figs. 2(a) and 2(b). We can observe the structural transitions of LiBH_4 upon heating and cooling in Fig. 2(a).^{13–15} The integrated diffraction intensities of LiCl shown in Fig. 2(b) clearly demonstrate the drastic decrease in the LiCl amount accompanied by the LT-HT transition at around 370 K upon heating and the increase by the HT-LT transition at around 330 K upon cooling, that is, LiCl partially dissolves into LiBH_4 and precipitates from it accompanied by the transition. From the results of the Rietveld analysis shown in Table I, it is found that the unit-cell volume of the LT phase measured at 296 K after heating is smaller than that of the as-milled (before heating) LT phase, indicating that a part of LiCl dissolved into the HT phase still remains in the LT phase [ionic radius: Cl^- (1.68 Å) < $[\text{BH}_4]^-$ (2.05 Å)],¹⁶ while the other XRD study reported the dissolution of LiCl in the HT phase only.¹⁷

The stabilization of the HT phase of $\text{LiBH}_4 + 0.33\text{LiCl}$ was also investigated by DSC detecting the structural transition as shown in Fig. 3.¹⁵ The enthalpy change of the transition in $\text{LiBH}_4 + 0.33\text{LiCl}$ is almost the same as that in pure and milled LiBH_4 within 2% differences. Therefore, the stabilization of the HT phase by the LiCl dissolution relates to

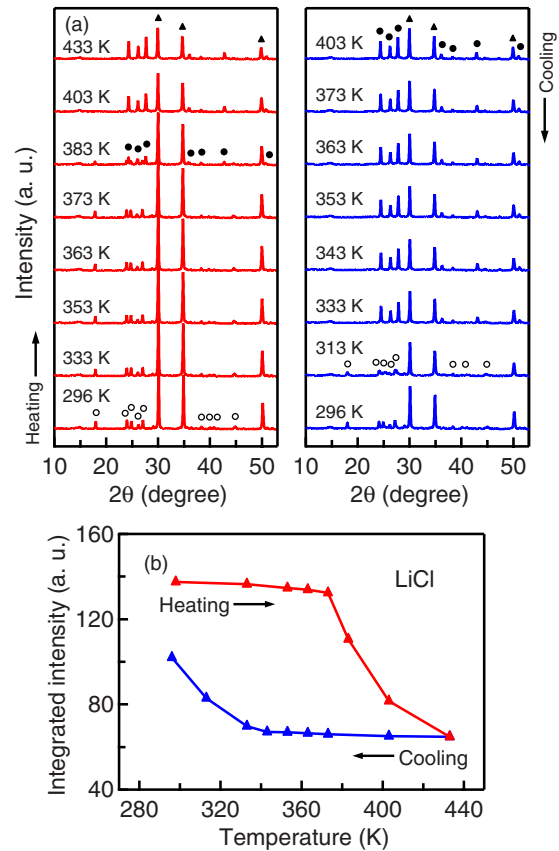


FIG. 2. (Color online) (a) XRD profiles (*in situ* measurements) of milled $\text{LiBH}_4 + 0.33\text{LiCl}$ and (b) temperature dependence of integrated diffracted intensity of LiCl in $\text{LiBH}_4 + 0.33\text{LiCl}$. Open and closed circles and triangle denote the LT and HT phases of LiBH_4 and LiCl, respectively. Red and blue data were taken in the heating and cooling runs, respectively. Heating and cooling rate was 5 K/min.

LiBH_4 .^{18–24} Recent studies have indicated that simultaneous rotations of $[\text{BH}_4]^-$ and “lattice anharmonicity” strongly affect the structural transition.^{22–24} Retaining of the anharmonicity due to the partial substitution of $[\text{BH}_4]^-$ by Cl^- might lead to the stabilization of the HT phase. It was also reported that alkali borohydrides with longer distances between neighboring $[\text{BH}_4]^-$ exhibit the lower transition temperatures.²⁵ In fact, we experimentally confirmed the room-temperature stabilization of the HT phase by the LiI dissolution with large anion I^- .²⁶

The enhanced electrical conductivity in the LT phase of milled $\text{LiBH}_4 + 0.33\text{LiCl}$ as compared to those of pure and

TABLE I. Lattice constant and unit-cell volume of the LT phases of milled $\text{LiBH}_4 + 0.33\text{LiCl}$ before (as-milled) and after heating, determined from *in situ* XRD measurement at 296 K shown in Fig. 2. Data for pure LiBH_4 are also shown for comparison. Smaller volume of milled $\text{LiBH}_4 + 0.33\text{LiCl}$ before heating than that of pure LiBH_4 implies that LiCl is already dissolved into LiBH_4 by the mechanical milling.

	Lattice constant (Å)			Unit-cell volume (Å ³)
	<i>a</i>	<i>b</i>	<i>c</i>	
Pure LiBH_4	7.184	4.440	6.807	217.1
Milled $\text{LiBH}_4 + 0.33\text{LiCl}$				
Before heating	7.167	4.426	6.802	215.8
After heating	7.142	4.401	6.802	213.8

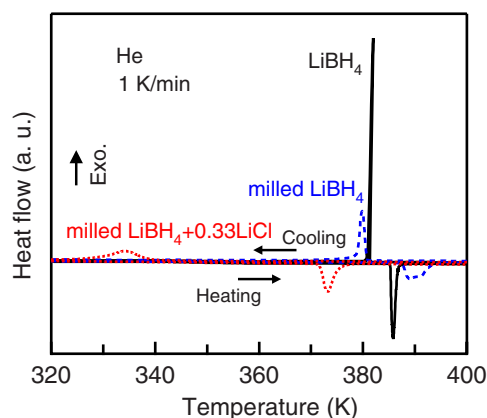


FIG. 3. (Color online) DSC curves (endothermic reactions for heating and exothermic ones for cooling) of pure LiBH_4 (black solid line), milled LiBH_4 (blue broken line) and milled $\text{LiBH}_4+0.33\text{LiCl}$ (red dotted line). Heating and cooling rate was 1 K/min.

milled LiBH_4 can be well retained after the tenth heating and cooling runs, as also shown in Fig. 1(b). The origin of the enhanced conductivity can probably be attributed to the higher mobility of the Li ions due to a modified unit-cell volume (decrease of 1.5% as shown in Table I) and/or to a different polarizability between $[\text{BH}_4]^-$ and Cl^- (Ref. 27) because activation energies for conduction in the LT phase decrease by the dissolution of LiCl [0.58 eV for $\text{LiBH}_4+0.33\text{LiCl}$, as opposed to 0.69 eV for pure LiBH_4 (Ref. 1)]. Another possibility might be an increase of carrier concentration caused by a dispersion effect of residual/precipitated LiCl.²⁸ The detailed studies are now underway and would greatly contribute to the development of solid electrolytes based on LiBH_4 and related hydrides²⁹ for solid-state batteries.

In summary, the effects of the addition of LiCl to LiBH_4 exhibiting lithium superionic conduction accompanied by structural transition at around 390 K were investigated. The higher electrical conductivity due to the mechanical milling effect is diminished after the transition. However, it is found that the LiCl addition drastically affects both the transition and conductivity of LiBH_4 . The transition from LT to HT (superionic conduction) phases is already observed at 370 K upon heating, and the HT phase can be retained at approximately 350 K (ac complex impedance measurement)-330 K (*in situ* XRD measurement) upon cooling. Hysteresis becomes 20–40 K by the LiCl addition (for $x=0.33$, in $\text{LiBH}_4+x\text{LiCl}$), as opposed to 4 K for pure and milled LiBH_4 . Further, the conductivity in the LT phase is more than one or two orders of magnitude higher than those of pure and milled LiBH_4 , and the enhanced conductivity in the LT phase can be retained after the tenth heating and cooling runs. These properties could be attributed to the dissolution of LiCl into LiBH_4 , suggested by *in situ* XRD measurement.

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- ¹M. Matsuo, Y. Nakamori, S. Orimo, H. Maekawa, and H. Takamura, *Appl. Phys. Lett.* **91**, 224103 (2007).
- ²Y. Nakamori, S. Orimo, and T. Tsutaoka, *Appl. Phys. Lett.* **88**, 112104 (2006).
- ³M. Matsuo, Y. Nakamori, K. Yamada, and S. Orimo, *Appl. Phys. Lett.* **90**, 232907 (2007).
- ⁴N. Machida, H. Maeda, H. Peng, and T. Shigematsu, *J. Electrochem. Soc.* **149**, A688 (2002).
- ⁵M. Tatsumisago, F. Mizuno, and A. Hayashi, *J. Power Sources* **159**, 193 (2006).
- ⁶N. Ohta, K. Takada, L. Zhang, R. Ma, M. Osada, and T. Sasaki, *Adv. Mater.* **18**, 2226 (2006).
- ⁷F. Berkemeier, M. S. Abouzari, and G. Schmitz, *Appl. Phys. Lett.* **90**, 113110 (2007).
- ⁸T. Kobayashi, Y. Imade, D. Shishihara, K. Homma, M. Nagao, R. Watanabe, T. Yokoi, A. Yamada, R. Kanno, and T. Tatsumi, *J. Power Sources* **182**, 621 (2008).
- ⁹C. R. C. The, in *Handbook of Solid State Electrochemistry*, edited by B. P. J. Gellings and H. J. M. Bouwmeester (CRC, Boca Raton, 1997), pp. 207–213.
- ¹⁰A. V. Deshpande and V. K. Deshpande, *Solid State Ionics* **154**, 433 (2002).
- ¹¹Y. Tomita, H. Matsushita, K. Kobayashi, Y. Maeda, and K. Yamada, *Solid State Ionics* **179**, 867 (2008).
- ¹²F. Izumi and T. Ikeda, *Mater. Sci. Forum* **321**, 198 (2000).
- ¹³J.-Ph. Soulié, G. Renaudin, R. Cerny, and K. Yvon, *J. Alloys Compd.* **346**, 200 (2002).
- ¹⁴A. Züttel, S. Rentsch, P. Fisher, P. Wenger, P. Sudan, Ph. Mauron, and Ch. Emmenegger, *J. Alloys Compd.* **356**, 515 (2003).
- ¹⁵Deviations of the transition temperatures between the impedance, DSC, and *in situ* XRD measurements were due to the different heating rates.
- ¹⁶*CRC Handbook of Chemistry and Physics*, 88th ed., edited by D. R. Lide (CRC, Boca Raton, 2007), pp. 12–27.
- ¹⁷L. Mosegaard, B. Møller, J. E. Jørgensen, Y. Filinchuk, Y. Cerenius, J. C. Hanson, E. Dimasi, F. Besenbacher, and T. R. Jensen, *J. Phys. Chem. C* **112**, 1299 (2008).
- ¹⁸Z. Łodziana and T. Vegge, *Phys. Rev. Lett.* **93**, 145501 (2004).
- ¹⁹H. Hagemann, S. Gomes, G. Renaudin, and K. Yvon, *J. Alloys Compd.* **363**, 126 (2004).
- ²⁰M. R. Hartman, J. J. Rush, T. J. Udovic, R. C. Bowman, Jr., and S.-J. Hwang, *J. Solid State Chem.* **180**, 1298 (2007).
- ²¹N. A. Zarkevich and D. D. Johnson, *Phys. Rev. Lett.* **100**, 040602 (2008).
- ²²Y. Filinchuk, D. Chernyshov, and R. Cerny, *J. Phys. Chem. C* **112**, 10579 (2008).
- ²³F. Buchter, Z. Łodziana, Ph. Mauron, A. Remhof, O. Friedrichs, A. Borgschulte, and A. Züttel, *Phys. Rev. B* **78**, 094302 (2008).
- ²⁴A.-M. Racu, J. Schoenes, Z. Łodziana, A. Borgschulte, and A. Züttel, and J. Phys. Chem. A **112**, 9716 (2008).
- ²⁵C. C. Stephenson, D. W. Rice, and W. H. Stockmayer, *J. Chem. Phys.* **23**, 1960 (1955).
- ²⁶H. Maekawa, M. Matsuo, H. Takamura, M. Ando, Y. Noda, T. Karahashi, and S. Orimo, *J. Am. Chem. Soc.* **131**, 894 (2009).
- ²⁷R. Kanno and M. Murayama, *J. Electrochem. Soc.* **148**, A742 (2001).
- ²⁸H. Maekawa, T. Iwatani, H. Shen, T. Yamamura, and J. Kawamura, *Solid State Ionics* **178**, 1637 (2008).
- ²⁹S. Orimo, Y. Nakamori, J. R. Eliseo, A. Züttel, and C. M. Jensen, *Chem. Rev.* **107**, 4111 (2007).